triOS COLLEGE
BUSINESS ◆ TECHNOLOGY ◆ HEALTHCARE

Course Syllabus

Course Code:	MWD2C (Swift)
Course Name:	Swift Development
Course Length:	80 hours (20 Days)

Course Description

In this course, students are introduced to object-oriented programming using the Swift programming language. More specifically, students will learn Swift expressions, collections, and types, as well as enumerations, protocols and generics.

Course Learning Outcomes

Upon successful completion of this course, students will be able to:

1. Identify the key elements of a Swift application.
2. Describe the object-oriented features of the Swift programming language.
3. Detail the procedures used to create and debug Swift applications.
4. Describe the common programming constructs that can be implemented within a Swift application.
5. Create a Swift application that solves a problem in a logical manner.
6. Use control structures and object-oriented features appropriately within a Swift application.
7. Implement custom types within a Swift application.
8. Test and debug an existing Swift application.

Materials and Resources

Textbooks:	Book 1: Wenderlich, R. (2023). Swift Apprentice: Fundamentals. Kodeco. Book 2: Wenderlich, R. (2023). Swift Apprentice: Beyond the Basics. Kodeco.
Software:	macOS (latest version), Apple Xcode, Git

Course Format

Lecture:

Knowledge transfer of concepts and practices will be done using PowerPoint presentations and/or the whiteboard, as well as software walkthroughs and demonstrations. Lectures go beyond a simple reiteration of the text or slides; information is contextualized with the use of examples. Topics and outcomes are the driving force of the lectures.

Discussion:
Points of discussion are interspersed throughout the course to keep you engaged with the material. Participation is encouraged, and the instructor is responsible for keeping discussions on track.

Activities/Exercises:
Short exercises will be used to explore concepts addressed in the lesson. Suggested extra activities are listed.

Supervised Lab:
Supervised lab activities will give students practical experience using the concepts and the software.

Class Preparation:
Review the class lectures to become familiar with the primary concepts before addressing them in-depth in class. Instructors assume you are prepared and will only cover the most important topics during class time.

Evaluation Breakdown

Assignments:

- Each Friday, students will complete a practical assignment that reflects the topics covered during the week. There are four assignments throughout the course that account for 90% of the final grade.

Professional Performance:

- Professional performance includes attendance, punctuality, participation in class, completion of all assignments, and class preparation. Professional performance is worth 10% of the final grade.

Evaluation Scale

A+	90–100 %	
A	80–89	
B+	75–79	
B	70–74	
B-	65–69	
C+	60–64	Pass Mark for triOS College
C	55–59	
C-	50–54	
D	40–49	
F	0–39	

Please Note: The pass mark for triOS College is 60%.

Students who miss tests or a final project submission due to medical reasons, and can provide a doctor's note, will be given a chance to resubmit at a later date.

The consequence of submitting a plagiarized, purchased, or in any manner inappropriately negotiated or falsified test, project, or any evaluated material, is a grade of zero on the material.

Daily Plan at a Glance

	DAY 1	DAY 2	DAY 3	DAY 4	DAY 5
Lecture	• Course Intro • Expressions, Variables, and Constants	• Types and Operations	• Control Flow	• Functions • Optionals	• N/A
Activity	• Book 1: Ch. 1 Challenges	• Book 1: Ch. 2 Challenges	• Book 1: Ch. 3 and 4 Challenges	• Book 1: Ch. 5 and 6 Challenges	• Assignment 1: Creating Temperature Functions
Reading	• Book 1: Ch. 1	• Book 1: Ch. 2	• Book 1: Ch. 3 and 4	• Book 1: Ch. 5 and 6	• N/A
	DAY 6	DAY 7	DAY 8	DAY 9	DAY 10
Lecture	• Arrays, Dictionaries, and Sets • Collection Iteration with Closures	• Strings • Regex	• Structures • Properties	• Methods • Classes	• N/A
Activity	• Book 1: Ch. 7 and 8 Challenges	• Book 1: Ch. 9 and 10 Challenges	• Book 1: Ch. 11 and 12 Challenges	• Book 1: Ch. 13 and 14 Challenges	• Assignment 2: Applying Arrays and Structs to a TV Series
Reading	• Book 1: Ch. 7 and 8	• Book 1: Ch. 9 and 10	• Book 1: Ch. 11 and 12	• Book 1: Ch. 13 and 14	• N/A

	DAY 11	DAY 12	DAY 13	DAY 14	DAY 15
Lecture	• Advanced Classes • Enumerations	• Protocols • Generics	• Access Control, Code Organization, and Testing • Custom Operators, Subscripts, and Keypaths	• Result Builders • Pattern Matching	• N/A
Activity	• Book 1: Ch. 15 & 16 Challenges	• Book 1: Ch. 17 and 18 Challenges	• Book 2: Ch. 1 and 2 Challenges	• Book 2: Ch. 3 and 4 Challenges	• Assignment 3: Working with Computer Classes
Reading	• Book 1: Ch. 15 & 16	• Book 1: Ch. 17 and 18	• Book 2: Ch. 1 and 2	• Book 2: Ch. 3 and 4	• N/A
	DAY 16	DAY 17	DAY 18	DAY 19	DAY 20
Lecture	• Error Handling • Encoding and Decoding Types	• Memory Management • Value and Reference Types	• Property Wrappers • Protocol-Oriented Programming	• Advanced Protocols and Generics • Concurrency	• N/A
Activity	• Book 2: Ch. 5 and 6 Challenges	• Book 2: Ch. 7 and 8 Challenges	• Book 2: Ch. 9 and 10 Challenges	• Book 2: Ch. 11 and 12 Challenges	• Assignment 4: Defining a Store for Digital Content
Reading	• Book 2: Ch. 5 and 6	• Book 2: Ch. 7 and 8	• Book 2: Ch. 9 and 10	• Book 2: Ch. 11 and 12	• N/A

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Daily Plan Summary

Day 1:

Topics:	• Course Introduction • Expressions, Variables, and Constants
Activities:	• Book 1: Ch. 1 Challenges
Readings:	• Book 1: Ch. 1

Day 2:

Topics:	• Types and Operations
Activities:	• Book 1: Ch. 2 Challenges
Readings:	• Book 1: Ch. 2

Day 3:

Topics:	• Control Flow
Activities:	• Book 1: Ch. 3 and 4 Challenges
Readings:	• Book 1: Ch. 3 and 4

Day 4:

Topics:	• Functions • Optionals
Activities:	• Book 1: Ch. 5 and 6 Challenges
Readings:	• Book 1: Ch. 5 and 6

Day 5:

Topics:	• Day 1–Day 4
Assignments:	• Assignment 1: Creating Temperature Functions

Day 6:

Topics:	• Arrays, Dictionaries, and Sets • Collection Iteration with Closures
Activities:	• Book 1: Ch. 7 and 8 Challenges
Readings:	• Book 1: Ch. 7 and 8

Day 7:

Topics:	• Strings • Regex
Activities:	• Book 1: Ch.9 and 10 Challenges
Readings:	• Book 1: Ch. 9 and 10

Day 8:

Topics:	• Structures • Properties
Activities:	• Book 1: Ch. 11 and 12 Challenges
Readings:	• Book 1: Ch. 11 and 12

Day 9:

Topics:	• Methods • Classes
Activities:	• Book 1: Ch. 13 and 14 Challenges
Readings:	• Book 1: Ch. 13 and 14

Day 10:

Topics:	• Day 6–Day 9
Assignments:	• Assignment 2: Applying Arrays and Structs to a TV Series

Day 11:

Topics:	• Advanced Classes • Enumerations
Activities:	• Book 1: Ch. 15 and 16 Challenges
Readings:	• Book 1: Ch. 15 and 16

Day 12:

Topics:	• Protocols • Generics
Activities:	• Book 1: Ch. 17 and 18 Challenges
Readings:	• Book 1: Ch. 17 and 18

Day 13:

Topics:	• Access Control, Code Organization, and Testing • Custom Operators, Subscripts, and Keypaths
Activities:	• Book 2: Ch. 1 and 2 Challenges
Readings:	• Book 2: Ch. 1 and 2

Day 14:

Topics:	• Result Builders • Pattern Matching
Activities:	• Book 2: Ch. 3 and 4 Challenges
Readings:	• Book 2: Ch. 3 and 4

Day 15:

Topics:	• Day 11–Day 14
Assignments:	• Assignment 3: Working with Computer Classes

Day 16:

Topics:	• Error Handling • Encoding and Decoding Types
Activities:	• Book 2: Ch. 5 and 6 Challenges
Readings:	• Book 2: Ch. 5 and 6

Day 17:

Topics:	• Memory Management • Value and Reference Types
Activities:	• Book 2: Ch. 7 and 8 Challenges
Readings:	• Book 2: Ch. 7 and 8

Day 18:

Topics:	• Property Wrappers • Protocol-Oriented Programming
Activities:	• Book 2: Ch. 9 and 10 Challenges
Readings:	• Book 2: Ch. 9 and 10

Day 19:

Topics:	• Advanced Protocols and Generics • Concurrency
Activities:	• Book 2: Ch. 11 and 12 Challenges
Readings:	• Book 2: Ch. 11 and 12

Day 20:

Topics:	• Day 16–Day 19
Assignments:	• Assignment 4: Defining a Store for Digital Content

Instructor and Student Role Expectations

College instructors are strongly committed to seeing students achieve the learning objectives of each course within a program.

As your instructor, I will:

- Encourage contact between students.
- Develop reciprocity and cooperation among students.
- Encourage active learning.
- Give prompt feedback.
- Emphasize time-on-task.
- Communicate high expectations.
- Respect diverse talents and ways of learning.

In return, it is expected that you will uphold certain values and behaviours, maintaining a productive learning environment for everyone in class.

As a student, you will:

- Arrive on time prepared to engage in the work of the class (for example, taking notes, participating in group discussions and activities, and so on.)
- Stay alert and participate throughout class.
- Avoid talking with your peers while the instructor or other students are speaking.
- Be courteous and act with decorum toward your peers and anyone who may be in class facilitating the learning process.
- Ask questions and participate in discussion but raise your hand before doing so.

- Submit assignments or other work on the date it is due. Exceptions are made only for extraordinary situations and with the approval of the instructor.
- Turn off cellphones, pagers, radios, and other such electronic devices during class. Exceptions are permitted only with the instructor's prior consent.

Policies and Procedures

For full detail, please see the Student Handbook.

Attendance Policy

Attendance is a mandatory requirement of all provincial education departments and the College. All students are expected to attend each scheduled class and are responsible for fulfilling course requirements they missed during an absence. It is the responsibility of the student to notify the College if they are going to be absent. This should be done prior to the start of class.

For specific policies and procedures regarding absences, attendance, and disciplinary measures, please refer to your Student Handbook.

Academic Integrity

Academic integrity is the core value at the College. The five values most often associated with academic integrity include:

- Honesty
- Trust
- Fairness
- Respect
- Responsibility

Academic integrity is the commitment to support these five values, even in the face of adversity. Just as your personal sense of integrity makes a statement about you as an individual, your attitude toward learning defines you.

Good students do not cheat or cut corners. They take responsibility for managing their own learning so that they become lifelong learners. The level of academic integrity a student demonstrates is measured by the amount of energy, effort, and focus that student is willing to put into their learning.

Academic dishonesty includes, but is not limited to:

- Cheating on assignments, quizzes, and exams by copying another student's work or by using unauthorized resources during a quiz or exam.
- Plagiarism — unauthorized use or close imitation of the language and thoughts of another author and misrepresenting the work as one's own. Development projects must not plagiarize code outside any frameworks and code-reuse must be approved by the instructor.

- Purchasing, selling, or sharing quizzes, exams, projects, and assignments.
- Use of unlicenced software.
- Talking during exams.

For specific policies and procedures regarding cheating, appealing grades, and disciplinary measures, please refer to your Student Handbook.